Online Appendix for Aging and Sectoral Productivity Gap

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A Proofs and Model Appendix

A.1 Derivation for Population Evolution

The evolution of population of the economy is characterized by the following transition parameter: (1) ω is the probability of remaining as a worker in the next period; (2) γ is the probability that a retiree will survive in the next period; (3) n is the population growth rate.

The average time for an individual to stay as a worker is given by

$$T_w = \sum_{t=0}^{\infty} (\omega)^t = \frac{1}{1-\omega}, \quad \omega \in (0,1).$$

The average retirement period for a retiree is given by

$$T_r = \sum_{t=0}^{\infty} (\gamma)^t = \frac{1}{1-\gamma}, \quad \gamma \in (0,1).$$

Let $\mathcal{R}_t$ be the total survival retirees in period t, recall the probability for a worker to become a retiree is $1-\omega$, so $\mathcal{R}_t$ is given by

$$\mathcal{R}_t = \sum_{j=1}^{\infty} \gamma^{j-1}(1-\omega)N_{t-j},$$

combine with population growth where

$$N_{t-j}(1+n)^j = N_t,$$

we obtain

$$\mathcal{R}_t = \sum_{j=1}^{\infty} \gamma^{j-1}(1-\omega)\frac{N_t}{(1+n)^j} = \left(\frac{1-\omega}{1+n}\right) N_t \sum_{t=0}^{\infty} \left(\frac{\gamma}{1+n}\right)^j,$$

thus $\mathcal{R}_t = N_t \left(\frac{1-\omega}{1+n-\omega}\right)$. The ratio of retirees to workers ψ is given by

$$\psi = \frac{\mathcal{R}_t}{N_t} = \frac{1-\omega}{1+n-\gamma}.$$

A.2 Proof for Proposition 1

Retiree's Marshallian demand function

$$c^r_{s,t} = \frac{\theta^r_s(e^r_t, p_{s,t})e^r_t}{p_{s,t}} = \left[\phi^r + \nu^r \left(\frac{p^{\phi^r}_{s,t}}{e^r_t}\right)^{\eta^r}\right] \cdot \frac{e^r_t}{p_{s,t}} = \phi^r \left(\frac{e^r_t}{p_{s,t}}\right) + \nu^r \frac{p^{\phi^r \eta^r - 1}_{s,t}}{(e^r_t)^{\eta^r - 1}}$$

and

$$c^r_{m,t} = (1-\theta^r_s)e^r_t = \left(1 - \phi^r - \nu^r \left(\frac{p^{\phi^r}_{s,t}}{e^r_t}\right)^{\eta^r}\right) e^r_t$$

Worker's Marshallian demand function

$$c^w_{i,s,t} = \frac{\theta^w_s(e^w_{i,t}, p_{s,t})e^w_{i,t}}{p_{s,t}} = \phi^w \left(\frac{e^w_{i,t}}{p_{s,t}}\right) + \nu^w \frac{p^{\phi^w \eta^w - 1}_{s,t}}{(e^w_{i,t})^{\eta^w - 1}}$$

and

$$c^w_{i,m,t} = (1-\theta^w_s)e^w_{i,t} = \left(1 - \phi^r - \nu^r \left(\frac{p^{\phi^w}_{s,t}}{e^w_{i,t}}\right)^{\eta^w}\right) e^w_{i,t}$$

Aggregate consumption

$$C_s(p_s, \psi) = \psi N \cdot c^r_s + N \int c^w_{i,s} dG = \psi N \cdot c^r_s + N\mathbb{E}[c^w_s]$$

and

$$C_m(p_s, \psi) = \psi N \cdot c^r_m + N \int c^w_{i,m} dG = \psi N \cdot c^r_m + N\mathbb{E}[c^w_m]$$

And efficient labor supply is given by

$$L_s(p_s) = \int_{\Omega_s(p_s)} z^i_s dG, \quad L_m(p_s) = \int_{\Omega_m(p_s)} z^i_m dG,$$

where $\Omega_s(p_s) = \{i : p_s z^i_s \geq z^i_m\}$, and $\Omega_m(p_s) = \{i : p_s z^i_s < z^i_m\}$

At Market Clearing we have

$$Y_s = C_s, \quad Y_m = C_m$$

So

$$D(p_s, \psi) := \frac{C_s(p_s, \psi)}{C_m(p_s, \psi)} - \frac{L_s(p_s)}{L_m(p_s)} = 0.$$

And IFT:

$$\frac{\partial p_s}{\partial \psi} = -\frac{D_\psi}{D_{p_s}}$$

We first calculate D_ψ

$$D_\psi = \frac{C_m \partial C_s/\partial\psi - C_s \partial C_m/\partial\psi}{C_m^2} = \frac{N}{C_m^2}\left(C_m c_s^r - C_s c_m^r\right)$$

Simplifying:

$$\begin{aligned}
D_\psi &= \frac{N}{C_m^2}\left[(\psi N c_m^r + N\mathbb{E}[c_m^w])c_s^r - (\psi N c_s^r + N\mathbb{E}[c_s^w])c_m^r\right] \\
&= \frac{N^2}{C_m^2}\left[(\psi c_m^r + \mathbb{E}[c_m^w])c_s^r - (\psi c_s^r + \mathbb{E}[c_s^w])c_m^r\right] \\
&= \frac{N^2}{C_m^2}\underbrace{\left(c_s^r\mathbb{E}[c_m^w] - c_m^r\mathbb{E}[c_s^w]\right)}_{\Delta}
\end{aligned}$$

So

$$\operatorname{sign}(D_\psi) = \operatorname{sign}\left(\frac{c_s^r}{c_m^r} - \frac{\mathbb{E}[c_s^w]}{\mathbb{E}[c_m^w]}\right)$$

It is convenient to see that $\frac{\partial}{\partial p_s}\left(\frac{C_s}{C_m}\right) < 0$, and from the previous cutoff rule, a increased p_s will lower the threshold $z_m/z_s \leq p_s$, more workers will enter service sector, thus $\frac{\partial}{\partial p_s}\left(\frac{L_s}{L_m}\right) > 0$, thus $D_p < 0$. By IFT,

$$\frac{dp_s}{d\psi} = -\frac{D_\psi}{D_p} > 0, \quad \text{iff} \quad \theta_s^r > \theta_s^w$$

Thus **Proposition 1** is proved.

A.3 Conditional Expectations

When $\alpha + \kappa = 0$, we have $\alpha = -\kappa$ ($\alpha < 0$), and

$$\ln z_s(q) = \bar{\alpha} - \kappa \ln q, \quad \ln z_m(q) = \bar{\alpha} - \kappa \ln(1-q).$$

For $q \sim U(0,1)$, assume q follows a unit uniform distribution,

$$E(\ln q | q \leq q_s) = \frac{1}{q_s}\int_0^{q_s} \ln q dq = \ln q_s - 1,$$

$$E(\ln(1-q_s)|q > q_s) = \frac{1}{1-q_s}\int_{q_s}^1 \ln(1-q)dq = \ln(1-q) - 1,$$

substitute the calculated expectations to the original expression of conditional expectations to obtain

$$E(\ln z_s|q \leq q_s) = \bar{\alpha} - \kappa \ln q_s + \kappa, \quad E(\ln z_m|q > q_s) = \bar{\alpha} - \kappa \ln(1 - q_s) + \kappa.$$

The derivation is similar for the case when $\alpha > 0$.

A.4 Steady State

At equilibrium, taking the exogenous population evolution process, economy-wide technology and individual productivity distribution $\{n, \omega, \psi, \gamma, \beta, \nu^w, \nu^r, \phi^r, \phi^w, \eta^r, \eta^w, R_t, X_t, G(z_s, z_m)\}$ as given, the sequence of endogenous state variables are$\{A_t^r, A_t^w, \mathbf{A}_t, N_t\}$ and control variables are$\{e_t^r, e_{i,t}^w, \mathbf{C}_{s,t}, \mathbf{C}_{m,t}, T_t, L_{s,t}, L_{m,t}, \mathbf{W}_t, \mathbf{Y}_{s,t}, \mathbf{Y}_{m,t}, p_{s,t}, N_{s,t}, N_{m,t}, \Omega_{s,t}, \Omega_{m,t}\}$.

The per capita steady state system is given by the below conditions:

A.4.1 Production

$$Y_s = X\bar{L}_s \tag{1}$$

$$Y_m = X\bar{L}_m \tag{2}$$

A.4.2 Wage rate

$$w_s = p_s X \tag{3}$$

$$w_m = X \tag{4}$$

A.4.3 Labor allocation

$$\frac{z_s^i}{z_m^i} \leq p_s \tag{5}$$

$$\Omega_s = \left\{ i | \frac{z_m^i}{z_s^i} \leq p_s \right\} \tag{6}$$

$$\Omega_m = \left\{ i | \frac{z_m^i}{z_s^i} > p_s \right\} \tag{7}$$

$$\bar{L}_s = \frac{\int_{i \in \Omega_s} z_s^i dG}{N} \tag{8}$$

$$\bar{L}_m = \frac{\int_{i \in \Omega_m} z_m^i dG}{N} \tag{9}$$

A.4.4 Consumption:

$$\theta_s^r(e^r, p_s) = \phi^r + \nu^r \left(\frac{p_s^{\phi^r}}{e^r}\right)^{\eta^r} \tag{10}$$

$$\theta_s^w(e_i^w, p_s) = \phi^w + \nu^w \left(\frac{p_s^{\phi^w}}{e_i^w}\right)^{\eta^w} \tag{11}$$

$$\bar{C}_s = \psi \frac{\theta_s^r e^r}{p_s} + \frac{\mathbb{E}[\theta_s^w e^w]}{p_s} \tag{12}$$

$$\bar{C}_m = \psi(1 - \theta_w^r)e^r + \mathbb{E}[(1 - \theta_s^w)e^w] \tag{13}$$

A.4.5 Wealth

$$\bar{W} = p_s X \bar{L}_s + X \bar{L}_m \tag{14}$$

$$e_r = (R/\gamma - 1)A_r \tag{15}$$

$$e_{i,w} = (R - 1)A_r + W_i \tag{16}$$

$$A = \psi N A_r + N \mathbb{E}[A_w] \tag{17}$$

A.5 Individual Asset

Define

$$a_t^r \equiv \frac{\mathbf{A}_t^r}{N_t}$$

and

$$N_{t+1} a_{t+1}^r = R_t N_t a_t^r + (1 - \omega) N_t \mathbb{E}[A_{t+1}^w]$$

divide N_t, we obtain

$$a_{t+1}^r = \frac{R_t}{1+n} a_t^r + \frac{1-\omega}{1+n} \mathbb{E}[A_{t+1}^w] \tag{18}$$

B Motivating Facts

B.1 Sectoral Classification of Aggregate Data

Table 1 provides our classification of service and manufacturing productivity based on the categories of Wolrd Bank Productivity project. The constructed productivity measures and productivity gaps are used in the motivating cross-country evidence and serve as the comparison benchmark.

Table 1: Sectoral Classification by World Bank Productivity

Sector name	**Description**
Agriculture:	
1. Agriculture	Agriculture, forestry, and fishing
Manufacturing:	
2. Mining	Mining and quarrying
3. Manufacturing	Manufacturing
4. Construction	Construction
Service:	
5. Utilities	Electricity, gas, steam and air conditioning supply
6. Trade services	Wholesale and retail trade; repair of motor vehicles and motorcycles; Accommodation and food service activities
7. Transport services	Transportation and storage; Information and communication
8. Financial and Business services	Financial and insurance activities; Real estate activities; Professional, scientific and technical activities; Administrative and support service activities
9. Other services	Public administration and defense; compulsory social security; Education; Human health and social work activities; Arts, entertainment and recreation; Other service activities; Activities of households as employers; undifferentiated goods- and services-producing activities of households for own use; Activities of extraterritorial organizations and bodies

Notes: This table displays the sectoral classification used in empirical analysis and calculation of sectoral productivity gap. For data annex, visit WB productivity project website: `https://www.worldbank.org/en/research/publication/global-productivity`

B.2 Expenditure Shares by Sector From CEX

Table 2 presents the classification of consumer-level expenditures in the CEX, following the classification schemes of Aguiar and Bils (2015) and Cravino et al. (2022). I divide the data into four time periods to illustrate the overall trend in household expenditure shares. The harmonized dataset is also used for estimating equation (3) and calibrating the parameters of the two-agent PIGL preference model.

Table 2: Expenditure shares by sector and period (%)

	Including housing				Excluding housing			
	82-91	92-01	02-11	12-23	82-91	92-01	02-11	12-23
Goods	**43.5**	**40.6**	**37.3**	**35.5**	**55.0**	**53.1**	**50.3**	**47.4**
Food at home	12.2	11.6	10.9	11.2	15.4	15.2	14.8	15.0
Vehicle	11.0	11.7	9.8	9.5	13.9	15.3	13.2	12.8
Gas	4.6	3.5	4.9	4.4	5.8	4.5	6.6	5.9
Entertainment equipment	3.8	3.9	4.0	3.6	4.7	5.1	5.4	4.8
Appliances	2.2	2.2	1.8	1.9	2.8	2.9	2.4	2.5
Clothing	3.1	2.4	1.5	0.9	4.0	3.1	2.1	1.2
Furniture & fixtures	2.1	1.6	1.4	1.4	2.6	2.2	1.8	1.9
Alcoholic beverages	1.3	1.0	0.9	1.0	1.7	1.3	1.2	1.3
Shoes & other apparel	1.3	1.0	0.7	0.6	1.7	1.3	1.0	0.7
Tobacco	1.1	0.9	0.8	0.6	1.3	1.2	1.0	0.8
Children's clothing	0.8	0.8	0.6	0.4	1.1	1.0	0.8	0.5
Services	**56.4**	**59.6**	**62.8**	**64.6**	**45.0**	**46.8**	**49.6**	**52.7**
Health	5.0	5.6	6.5	8.6	6.3	7.4	8.7	11.6
Utilities	7.9	7.7	8.2	7.8	10.0	10.1	11.1	10.4
Cash contributions	2.6	2.7	3.2	3.3	3.2	3.5	4.3	4.4
Car maintenance	4.3	4.5	3.9	4.0	5.5	5.8	5.2	5.4
Food away from home	5.0	4.5	4.8	5.1	6.3	5.8	6.5	6.8
Domestic services	3.2	3.2	3.3	3.7	4.0	4.3	4.5	5.0
Education	1.6	1.8	2.3	2.3	2.0	2.3	3.1	3.2
Entertainment	2.3	2.2	1.9	1.5	2.9	2.9	2.5	2.1
Public transport	1.3	1.4	1.2	1.3	1.7	1.8	1.6	1.8
Personal care	1.0	0.9	0.7	0.7	1.3	1.2	0.9	0.9
Housing	20.8	23.8	25.9	25.5	.	.	.	.
Personal insurance	1.4	1.3	0.9	0.8	1.8	1.7	1.2	1.1

Notes: This table displays the categorized expenditure share of household sample in the CEX data across periods. Calculations are based on the classification rule of Aguiar and Bils (2015) and Cravino et al. (2022).

Figure 1 displays the expenditure share of services by age group across different time periods, while Figure 2 categorizes the expenditure share by income quartiles. Both figures convey a similar message to the coefficient plot in the paper: the share of service expenditures has been increasing over time, reflecting consumers' shifting preferences and the ongoing expansion of the service sector.

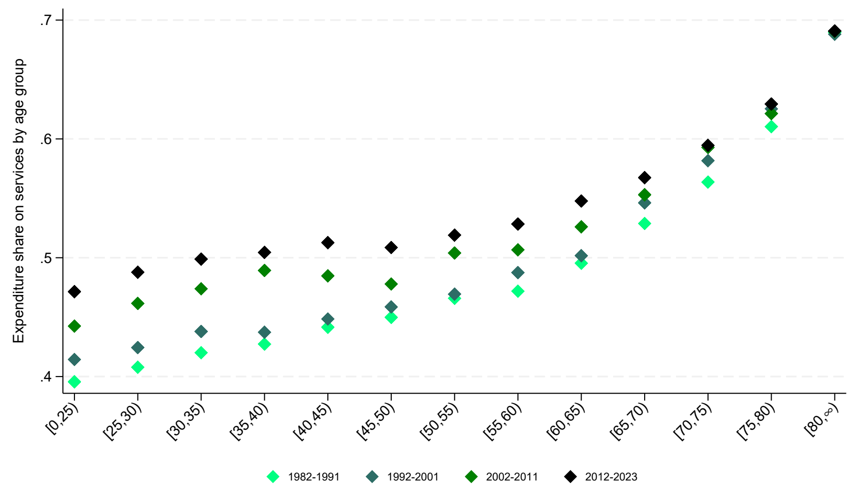

Figure 1: Expenditure Share of Service by Age Groups Across Time Periods

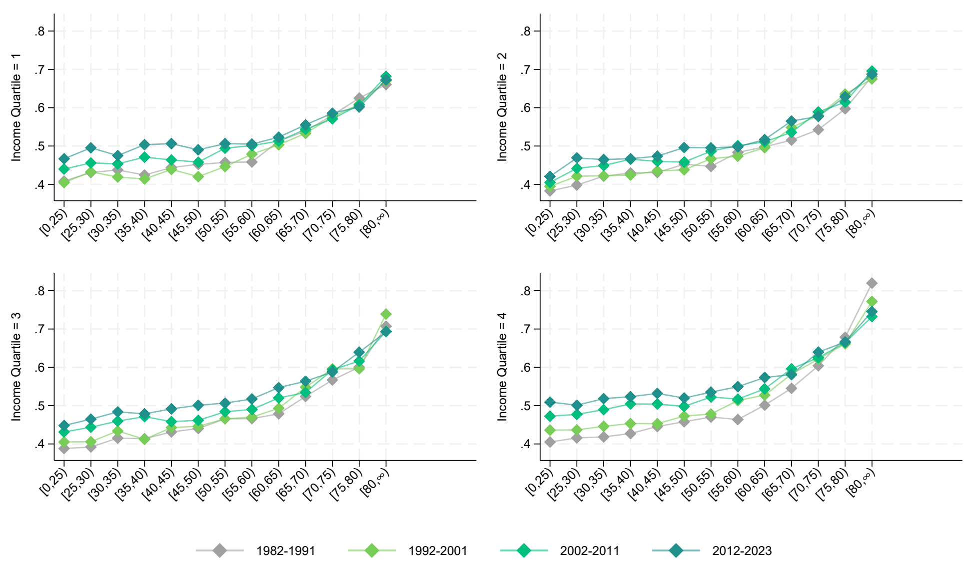

Figure 2: Expenditure Share of Service by Age Groups and Income Quartile

B.3 Calibration

Table 3: CPS Sectoral Classification

2 digit Code	Industry
Manufacturing	
2	Mining
3	Construction
4	Durable goods manufacturing
5	Non-durable goods manufacturing
Service	
6	Wholesale trade
7	Retail trade
8	Transportation and warehousing
9	Utilities
10	Information
11	Finance and insurance
12	Real estate and rental and leasing
13	Professional, scientific, and technical services
14	Management, administrative support, and waste management
15	Educational services
16	Health care and social assistance
17	Arts, entertainment, and recreation
18	Accommodation and food services
19	Private households
20	Other services, except private households
21	Public administration

Notes: This table display the classification of service and manufacturing occupation from CPS data when calibrating the dispersion level of the skill distribution and the correlation in Frank Copula.

C Quantitative Results

Table 4: Predicted Dynamics of Selected Countries

2000-2017	Productivity Gap	Va. Share	Emp. Share
United States	-0.044	0.259	0.028
	(-0.122)	(0.289)	(0.049)
United Kingdom	-0.043	0.026	0.027
	(-0.066)	(0.062)	(0.074)
Japan	-0.164	0.101	0.101
	(-0.382)	(0.001)	(0.086)
Germany	-0.080	0.0489	0.0509
	(-0.387)	(8.94e-07)	(0.080)
China	-0.371	0.144	0.175
	(-0.086)	(0.063)	(0.162)
India	-0.767	0.173	0.217
	(0.309)	(0.090)	(0.072)

Notes: This table compares the model-predicted dynamics of the productivity gap, value-added share, and employment share of services with the corresponding observed data. The dynamics are measured as the difference between the last and first year values. Actual data are reported in parentheses.

Figures 3 to 5 plot the coefficients obtained from regressions of the predicted values from benchmark model on the actual data. These results show that our benchmark model performs well in predicting the value-added share and employment share of services during the 2000–2017 period. However, it generates accurate predictions for productivity gaps only starting from 2014, and overall the model accounts for only about 20% of the observed productivity gap. And Figure 6 to figure 8 plot the coefficient obtained from regression of the model when demographics is the only driver. Figure 6 show that when countries only differ in demographics structure, the model cannot provide correct prediction on the level of productivity gaps, and only account for within country dynamics.

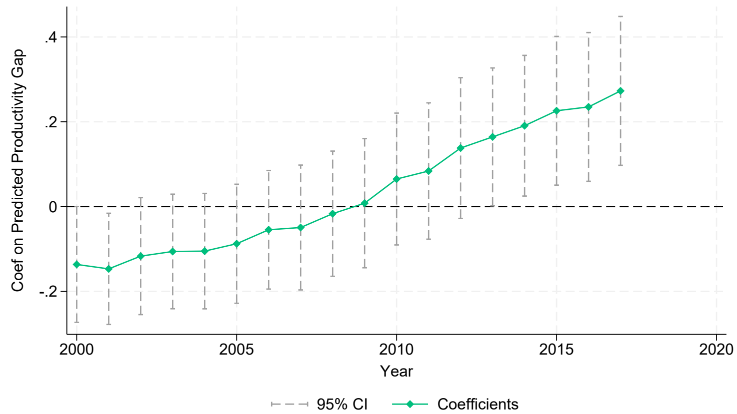

Figure 3: Benchmark Dynamics: Productivity Gap

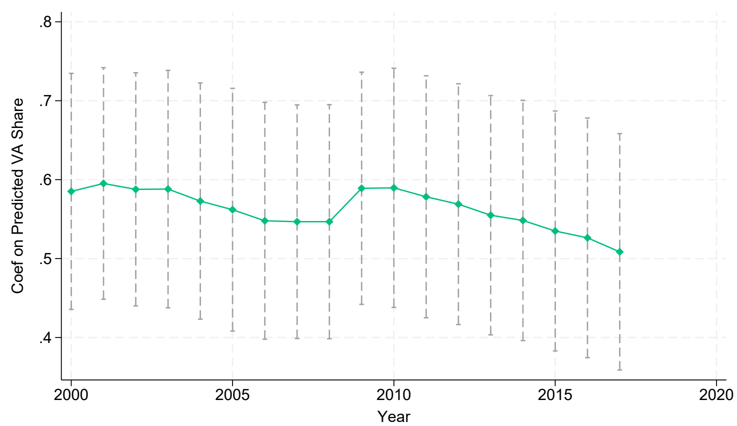

Figure 4: Benchmark Dynamics: Value Added Share

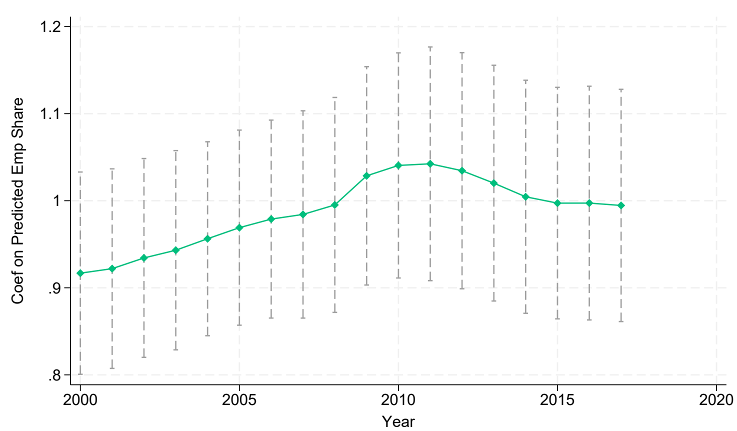

Figure 5: Benchmark Dynamics: Employment Share Share

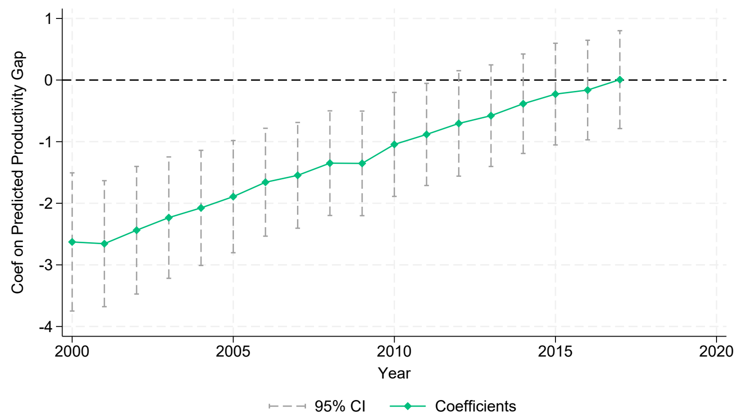

Figure 6: Demographics Only Dynamics: Productivity Gap

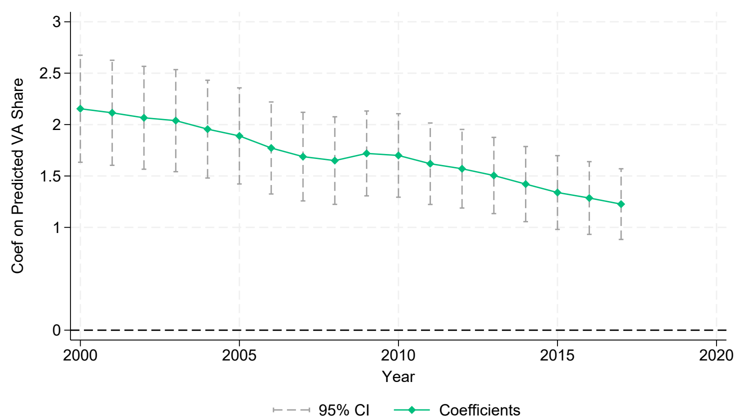

Figure 7: Demographics Only Dynamics: Value Added Share

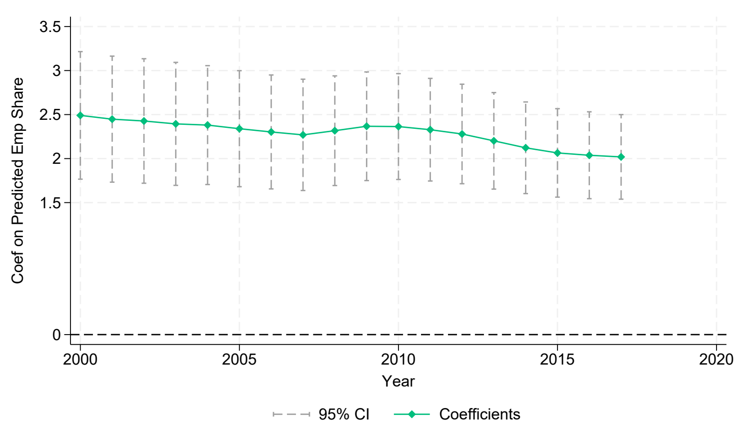

Figure 8: Demographics Only Dynamics: Employment Share Share

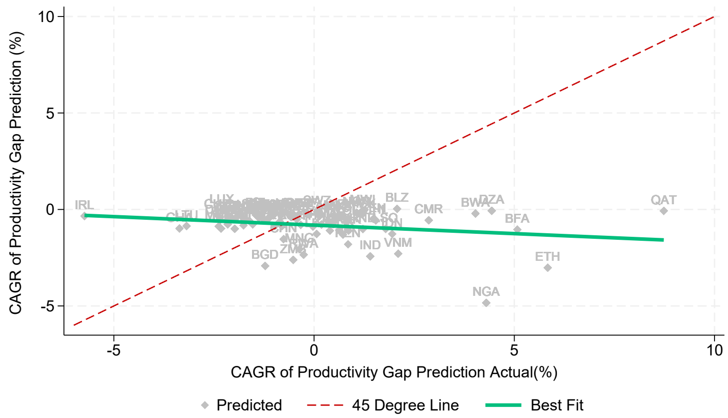

Figure 9: Within Country Dynamics: Productivity Gap

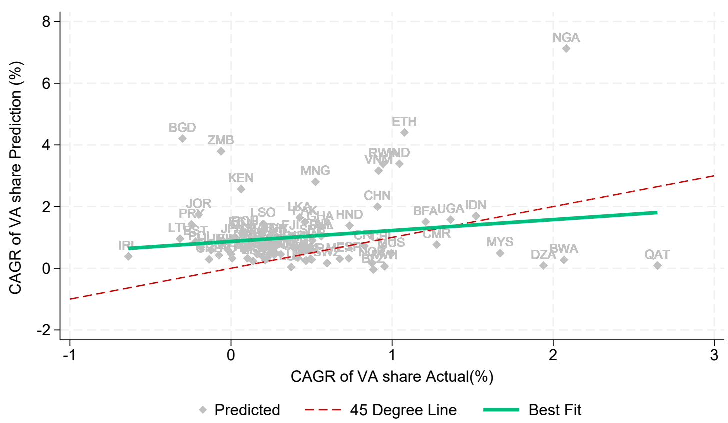

Figure 10: Within Country Dynamics: Value Added Share

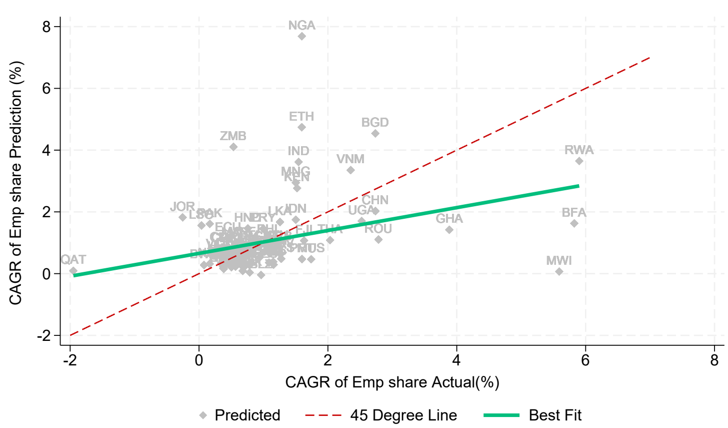

Figure 11: Within Country Dynamics: Employment Share Share

D Model Extension with Skill Types and Automation

I now extend our baseline model to incorporate automation as a mechanism for labor replacement, and to allow for differentiated roles of high- and low-skill workers in production, following the framework of Buera et al. (2022). In the benchmark model, overall technological progress is introduced exogenously and lies outside the production function, implying that automation–labor substitution is neglected. While the benchmark specification shows that using GDP per capita to approximate technological progress absorbs much of the dynamic effect of aging, the extended model allows us to explicitly identify and analyze the automation channel through which aging may affect sectoral productivity and labor allocation. By doing so, we can amplify the selection effect of aging without additional modification to the skill distribution.

D.1 Labor Force

Compared to the benchmark, workers are now classified into high-skill and low-skill types to specify the contribution of different types of worker in the production. And each worker now have two components in their skill levels z^i_{kj}: general skill level κ_k and sector-specific skill. Similarly, the sector-specific skills are drawn from a joint Fréchet distribution $F(a_{ks}, a_{km}) = C[F(a_{ks}), F(a_{km})]$ governed by a Frank Copula.

$$F(a_{ks}) = e^{-a_{ks}^{-\theta_{ks}}} \quad \text{and} \quad F(a_{km}) = e^{-a_{km}^{-\theta_{km}}} \quad k = H, L.$$

Whether a worker has high or low general skills is decided by a binary distribution where the probability of worker to be high-skilled is π. Normalizing the general skill level of low-skill level worker equal to 1, we have

$$z^i_{Hj} = \kappa \cdot a^i_{Hj}, \quad z^i_{Lj} = a^i_{Lj} \quad j = s, m.$$

D.2 Production

In the extended model, we assume the production of service and manufacturing share similar CES form, but in manufacturing, the production is given by a mixture of machine and labor. The production technologies for service and manufacturing are give by

$$Y_j = A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1-\alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{\sigma}{\sigma-1}}, \quad j = s, m$$

and

$$X_j = \left[\mu_j L_j^{\frac{\xi-1}{\xi}} + (1-\mu_j)(A_{jK} K_j)^{\frac{\xi-1}{\xi}} \right]^{\frac{\xi}{\xi-1}},$$

where Y_s is the output of service sector and Y_m is the output of manufacturing sector, H_j is the aggregate efficient labor of high skill worker and L_j is the aggregate efficient labor of low skill worker. X_j is the sector-specific production task composite, which is intuitively relative codifiable thus can be completed by both low skill workers and machine. In the benchmark extension we follow Buera et al. (2022) and assume the elasticity of substitution between high-skill labor and task composite σ is the same in both sectors; the elasticity of substitution between labor and capital ξ is also the same. And the efficient labor aggregators are given by

$$H_j = \int z_j^i dG(z_{H,s}, z_{H,m}), \quad j = s, m$$

and

$$L_j = \int z_j^i dG(z_{L,s}, z_{L,s}), \quad j = s, m.$$

D.3 Equilibrium

D.3.1 Profit Maximization

The representative firm in sector j solves the profit maximization problem:

$$\max_{H_j, K_j, L_j} \quad \Pi_j = p_j Y_j - w_H H_j - R_K K_j - w_L L_j. \quad \text{s.t.} \quad X_j \leq F(H, A)$$

From the FOCs we have

$$\begin{aligned}
w_H &= p_j A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1-\alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \alpha_j H_j^{-\frac{1}{\sigma}}, \\
w_L &= p_j A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1-\alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \cdot \left(1-\alpha_j\right) \mu_j X_j^{\frac{1}{\xi}-\frac{1}{\sigma}} \cdot L_j^{-\frac{1}{\xi}} \\
R_K &= p_j A_j \left[\alpha_j H_j^{\frac{\sigma-1}{\sigma}} + (1-\alpha_j) X_j^{\frac{\sigma-1}{\sigma}} \right]^{\frac{1}{\sigma-1}} \cdot \left(1-\alpha_j\right) \left(1-\mu_j\right) X_j^{\frac{1}{\xi}-\frac{1}{\sigma}} A_{jK} \cdot \left(A_{jK} K_j\right)^{-\frac{1}{\xi}}.
\end{aligned}$$

Rearranging the FOCs to obtain the optimized input ratio :

$$\frac{K_j}{L_j} = A_{jK}^{\xi-1} \left(\frac{1-\mu_j}{\mu_j} \right)^{\xi} \left(\frac{w_L}{R_K} \right)^{\xi}. \tag{19}$$

$$\frac{H_j}{X_j} = \left(\frac{\alpha_j}{1-\alpha_j} \right)^{\sigma} \left(\frac{c_{X,j}}{w_H} \right)^{\sigma} \tag{20}$$

D.3.2 Price Allocations

Consider the duality of profit maximization, start from the inner composite production CES:

$$X_j = \left[\mu_j L_j^{\frac{\xi-1}{\xi}} + (1-\mu_j)(A_{jK}K_j)^{\frac{\xi-1}{\xi}}\right]^{\frac{\xi}{\xi-1}},$$

and

$$\min_{L_j, K_j} w_L L_j + R_k K_j \quad \text{s.t.} \quad X_j = 1.$$

Define effective capital as $\tilde{K}_j \equiv A_{jK}K_j$, whose rental rate is $\tilde{R}_K \equiv R_K / A_{jK}$. The unit cost of X_j is given by standard CES form

$$c_{jX}(w_L, R_K) = \left[\mu_j^{\xi} w_L^{1-\xi} + (1-\mu_j)^{\xi}\left(\frac{R_K}{A_{jK}}\right)^{1-\xi}\right]^{\frac{1}{1-\xi}}. \tag{21}$$

Similarly for the outer CES we have

$$c_{jY}(w_H, w_L, R_K) = \frac{1}{A_j}\left[\alpha_j^{\sigma} w_H^{1-\sigma} + (1-\alpha_j)^{\sigma} c_{jX}(w_L, R_K)^{1-\sigma}\right]^{\frac{1}{1-\sigma}}. \tag{22}$$

Normalizing the wage of low-skill labor as 1, we obtain the nominal price of sector j.

$$\hat{\mathbf{p}}_j(w_H, R_K) = \frac{1}{A_j}\left\{\frac{\alpha_j}{w_H^{\sigma-1}} + (1-\alpha_j)\left[\mu_j + (1-\mu_j)\left(\frac{R_K}{A_{jK}}\right)^{1-\xi}\right]^{\frac{1-\sigma}{1-\xi}}\right\}^{\frac{1}{1-\sigma}}. \tag{23}$$

D.3.3 Labor Intensity in Production

The share of high-skill labor in sector j is given by $\vartheta_{jH} = \frac{w_H H_j}{w_H H_j + w_L L_j + R_K K_j}$, from the profit maximization conditions we have

$$\vartheta_{jH} = \left[1 + \frac{1-\alpha_j}{\alpha_j}\left(\frac{X_j}{H_j}\right)^{\frac{\sigma-1}{\sigma}}\right]^{-1},$$

similarly, the share of low-skill labor in sector j is given by

$$\vartheta_{jL} = \mu_j\left(\frac{L_j}{X_j}\right)^{\frac{\xi-1}{\xi}}\left[1 + \frac{\alpha_j}{1-\alpha_j}\left(\frac{H_j}{X_j}\right)^{\frac{\sigma-1}{\sigma}}\right]^{-1}.$$

D.4 Demand Function

Define

$$f_{jX} = \mu_j^{\xi} w_L^{1-\xi} + (1-\mu_j)^{\xi} \left(\frac{R_K}{A_{jK}} \right)^{1-\xi},$$

thus $c_{jX} = (f_{jX})^{\frac{1}{1-\xi}}$. Similarly define

$$f_{jY} = \frac{\alpha_j^{\sigma}}{w_H^{\sigma-1}} + (1-\alpha_j)^{\sigma} \left[\mu_j + (1-\mu_j) \left(\frac{R_K}{A_{jK}} \right)^{1-\xi} \right]^{\frac{1-\sigma}{1-\xi}},$$

thus $c_{jY} = \frac{f_{jY}^{\frac{1}{1-\sigma}}}{A_j}$, and the zero profit condition is

$$f_{jY} = (A_j \hat{\mathbf{p}}_j)^{1-\sigma}$$

In the Inner CES, the cost shares of low-skilled labor and capital are given by

$$s_{jL} \equiv \frac{w_L L_j}{c_{jX} X_j} = \frac{\mu_j^{\xi} w_L^{1-\xi}}{f_{jX}}, \quad s_{jK} \equiv \frac{R_K K_j}{c_{jX} X_j} = \frac{(1-\mu_j)^{\xi} \left(\frac{R_K}{A_{jK}} \right)^{1-\xi}}{f_{jX}}.$$

In the outer CES, the cost shares of high-skilled labor and production composite are given by

$$s_{jH} \equiv \frac{w_H H_j}{c_{jY} Y_j} = \frac{\alpha_j^{\sigma} w_H^{1-\sigma}}{f_{jY}}, \quad s_{jX} \equiv \frac{c_{jX} X_j}{c_{jY} Y_j} = \frac{(1-\alpha_j)^{\sigma} c_{jX}^{1-\sigma}}{f_{jY}}.$$

Rearrange the above expression of input shares, we obtain the demand function:

$$H_j = \frac{s_{jH} c_{jY} Y_j}{w_H} = \frac{s_{jH} \hat{\mathbf{p}}_j Y_j}{w_H}, \quad X_j = \frac{s_{jX} c_{jY} Y_j}{c_{jX}} = \frac{s_{jX} \hat{\mathbf{p}}_j Y_j}{c_{jX}}$$

and

$$L_j = \frac{s_{jL} s_{jX} \hat{\mathbf{p}}_j Y_j}{w_L}, \quad K_j = \frac{s_{jK} s_{jX} \hat{\mathbf{p}}_j Y_j}{R_K}.$$

And the demand functions are given by

$$H_j = \left[\frac{\alpha A_j \hat{\mathbf{p}}_j (w_H, R_K)}{w_H} \right]^{\sigma} \frac{Y_j}{A_j}, \tag{24}$$

$$L_j = \left[\frac{(1-\alpha_j) A_j \hat{\mathbf{p}}_j (w_H, R_K)}{c_{jX}} \right]^{\sigma} (\mu_j c_{jX})^{\xi} \frac{Y_j}{A_j}, \tag{25}$$

$$K_j = \left[\frac{(1-\alpha_j)A_j\hat{\mathbf{p}}_j(w_H, R_K)}{c_{jX}}\right]^{\sigma}\left[\frac{(1-\mu_j)A_{jK}c_{jX}}{R_K}\right]^{\xi}\frac{Y_j}{A_jA_{jK}}. \tag{26}$$

D.4.1 Selection

Workers choose the sector with highest pay and enter service sector if

$$\frac{z^i_{mH}}{z^i_{sH}} \leq \frac{\hat{\mathbf{p}}_s}{\hat{\mathbf{p}}_m}, \quad \frac{z^i_{mL}}{z^i_{sL}} \leq \frac{\hat{\mathbf{p}}_s}{\hat{\mathbf{p}}_m}$$

where the price of manufacturing sector is normalized to 1 as previous section.

D.5 Calibration

Skill Distribution

Using the same CPS dataset spanning from 2003 to 2018, we classified workers with college degree or above as High-Skill workers, and others are classified as Low-Skill worker. The classification in CPS follows: 1. Less Than High School (LTHS); 2. High School; 3. Some College (No degree) 4. College; 5. Advanced (Master or Above). Recalibrating the skill-distribution for High-Skill and Low-Skill workers, we obtain two sets of distribution parameters as well as skill premium and share of high-skilled labor in Table 5.

Table 5: Skill-Specific Distribution Parameters

π	κ	θ_{Hs}	θ_{Hm}	ρ_H	θ_{Ls}	θ_{Lm}	ρ_L
0.37	1.17	2.70	3.59	3.76	3.49	4.24	3.76

Technology Parameters

Given the cost share of high-skilled labor

$$s^H_j = \frac{w_H H_j}{w_H H_j + c_{Xj}X_j},$$

given equation (19), we obtain

$$\frac{s^H_j}{1-s^H_j} = \frac{w_H}{c_{X,j}}\left(\frac{\alpha_j}{1-\alpha_j}\right)^{\sigma}\left(\frac{c_{X,j}}{w_H}\right)^{\sigma} = \left(\frac{\alpha_j}{1-\alpha_j}\right)^{\sigma}\left(\frac{w_H}{c_{X,j}}\right)^{1-\sigma}.$$

Take logs we have a estimation equation

$$\log \frac{\hat{s}_j^H}{1-\hat{s}_j^H} = \sigma \log \frac{\alpha_j}{1-\alpha_j} + (1-\sigma) \log \frac{w_H}{c_{Xj}}, \tag{27}$$

where $\hat{s}_j^H$ is the empirical wage bill share. By estimating the above equation, we will have $1-\sigma$ as the estimated coefficient.

Given the cost minimization results, the share of high-skilled labor cost can be expressed as

$$\hat{s}_j^H = \frac{\alpha_j^\sigma w_H^{1-\sigma}}{\alpha_j^\sigma w_H^{1-\sigma} + (1-\alpha_j)^\sigma c_{Xj}^{1-\sigma}}, \tag{28}$$

for a given σ, this expression defines a one-to-one mapping between the input share parameter α_j and the high-skilled wage bill share, and we therefore calibrate α_j by inverting the equation to match the wage bill share. Turning to the inner CES, I set elasticity of substitution between low-skilled labor and capital to $\xi = 1.45$, following the estimates in Hubmer (2023). This value lies close to the upper bound of Hubmer's estimated range and reflects our modeling assumption that low-skilled labor is more readily substitutable by capital than labor on average.

Similarly, the input share of low-skilled labor can also be calibrated by inverting the below equation for a given ξ.

$$\hat{s}_j^L = \frac{\mu_j^\xi w_{Lj}^{1-\xi}}{\mu_j^\xi w_{Lj}^{1-\xi} + (1-\mu_j)^\xi (R_K/A_{jK})^{1-\xi}}. \tag{29}$$

Inverting the above equation give us

$$q_{jt} \equiv \left(\frac{R_{kt}/A_{jKt}}{w_{Ljt}}\right) = \left[\frac{\mu_j^\xi}{(1-\mu_j)^\xi} \cdot \frac{1-\hat{s}_{jt}^L}{\hat{s}_{jt}^L}\right]^{\frac{1}{1-\xi}}. \tag{30}$$

Since we do not observe the user cost of capital at the sector level, we exploit the scale invariance of the CES aggregator and normalize the relative effective price of capital to low-skilled labor in the base year. I set $\left(\frac{R_{kt}/A_{jKt}}{w_{Ljt}}\right) = 1$ and recover subsequent changes in relative factor costs from observed movements in factor cost shares. I aggregate industry-level compensation data from BEA-KLEMS to generate the empirical share of low-skilled worker wage bill. When using different benchmark years, the calibrated input share of low-skilled labor exhibits moderate variation. Specifically, the implied values of the low-skilled input share range from 0.39 to 0.47 in services and from 0.45 to 0.52 in manufacturing when taking 1997 and 2017 as benchmark years. This variation reflects both long-run changes in task composition and the normalization of relative factor prices in the base year. In our baseline calibration, I therefore set $\mu_s = 0.43$ and $\mu_m = 0.49$, corresponding to the midpoint of the estimated range.

From the unit cost of composite production from equation 21, rearranging the terms we can express c_{jX} in terms of wage rate and q_{jt}

$$c_{jXt} = w_L \cdot \left[\mu_j^{\xi} + (1-\mu_j)^{\xi} q_{jt}^{1-\xi}\right]^{\frac{1}{1-\xi}}.$$

Table 6: Technology Parameters

α_s	α_m	σ	μ_s	μ_m	ξ
			0.43	0.49	1.45

D.6 Results

References

Mark Aguiar and Mark Bils. Has consumption inequality mirrored income inequality? *American Economic Review*, 105(9):2725–2756, 2015.

Francisco J Buera, Joseph P Kaboski, Richard Rogerson, and Juan I Vizcaino. Skill-biased structural change. *The Review of Economic Studies*, 89(2):592–625, 2022.

Javier Cravino, Andrei Levchenko, and Marco Rojas. Population aging and structural transformation. *American Economic Journal: Macroeconomics*, 14(4):479–498, 2022.

Joachim Hubmer. The race between preferences and technology. *Econometrica*, 91(1):227–261, 2023.